



Haptic In-Vehicle Gesture Controls

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Abstract

Recent efforts have proposed the use of hand gestures to control in-vehicle infotainment systems (IVISs) in an attempt to reduce visual demand and therefore reduce crash risk. These efforts however lack tactile feedback resulting in a decreased sense of control over the user's intended actions. Here, we describe a demo that uses commercially available novel technologies which mitigate this problem by using focused ultrasound to accurately deliver a radiation force onto the user's operating hand. This new paradigm of mid-air haptic feedback presents new opportunities and challenges towards designing effective interaction languages for IVISs. In this paper, we describe our demo prototype and how it addresses some of these challenges.

Author Keywords

In-vehicle controls; eyes-free; mid-air haptics; gestures; demo.

CCS Concepts

• **Human-centered computing~Haptic devices** • *Human-centered computing~Gestural input* • *Human-centered computing~Interface design prototyping*

Introduction

A major trend in the automotive industry is the integration of increasingly large touchscreen displays for in-vehicle infotainment systems (IVISs). Touchscreen use in vehicles has however been shown to increase crash risks [6, 7, 11]. Moreover, their

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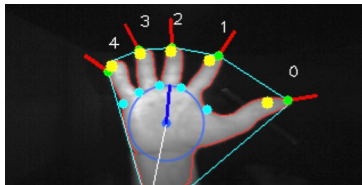


Figure 1: LEAP Motion mono-chromatic image with markers.

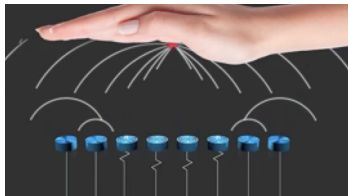


Figure 2: Creation of a focal point on the palm using an array of ultrasound transducers.

usability can be notably undermined due to the vibrations and lateral accelerations occurring while the vehicle is in motion, especially on non-ideal road conditions [1]. Much research has therefore focused on reducing the visual demand of IVISs both from a safety and user experience perspective [5, 6].

One promising solution is that of using hand gestures (HGs) [10, 13, 9] measured using infrared or computer vision devices like the LEAP Motion and Microsoft Kinect or other proprietary machine vision solutions. The primary benefit being that advanced hand-tracking-algorithms enable the design of IVISs that make use of the driver's peripheral vision, thereby decreasing visual demand, and consequently, visual distraction from the road [4]. The primary disadvantage of pure HG interaction control is the loss of tactile feedback, a key ingredient towards the sense of agency (SoA) - the subjective experience of voluntary control over your actions [3]. Speech and non-speech audio feedback have been proposed to offset this lack of tactile feedback [9]. However, these methods are not as effective and also interfere with other audio signals and external noise (e.g. due to an open door or window).

Very recent advancements in the manipulation and use of focused ultrasound as a non-invasive method to stimulate neuroreceptor structures in various parts of the hand have successfully addressed the SoA problem in HGs [3, 8]. In this extended abstract, we describe the world's first prototype demo (see Figure 3) which uses this new ultrasonic technology to provide *haptified* HGs for IVISs. We will limit our focus to the demo's physical and user interface design, and defer further in-depth technical and user studies for future work.

All the hardware and software used for building this demo are currently available to the consumer online.

Key Technologies Used in the Demo

Hand Gestures

There are many ways of using HGs for IVISs: pre-emptive gestures, function associated gestures, context sensitive gestures, global shortcut gestures, and natural dialogue gestures [12]. However, when designing HGs for IVIS, one is always restricted by the capabilities of the machine vision solution employed. In our demo, we have used a LEAP Motion controller (see Figure 1) - a small USB peripheral device that uses two monochromatic infrared cameras, three infrared LEDs, and machine vision proprietary software to track the user's hand and fingers. Its field of view is 150 degrees wide and 120 degrees deep with an effective range of approximately 60cm. The LEAP is particularly good at tracking the user's palm position, orientation, posture, and motion. We have therefore focused on these properties and designed our IVIS HGs that build on the LEAP's strengths. These are described in the next section.

Ultrasonic Haptic Feedback

A phased array composed of multiple ultrasonic transducers can be used to focus ultrasound at a specified location in three dimensions. A schematic representation is shown in Figure 2. The (x,y,z) coordinates for creating the focal point (e.g., at the center of the user's palm or fingertip) are obtained from the LEAP Motion API. For a tactile sensation to be perceived however, an acoustic radiation force must be induced. This is achieved by amplitude modulating the carrier waves at 200 Hz as to create shear waves in the skin tissue (i.e., a small localized skin displacement), which is perceived by the user as haptic feedback. In

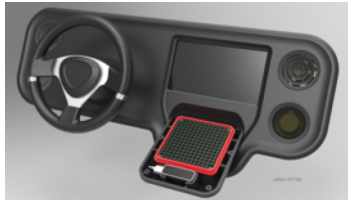


Figure 3: Graphics and photograph of the demo setup showing the position of the LEAP Motion, the Ultrahaptics device, an 11-inch monitor and the user's hand, relative to the steering.



Figure 4: Displayed screen on the monitor indicates the level of the audio volume and the fan.

our demo, we have used the Ultrahaptics [14] TOUCH[®] Development Kit which has $14 \times 14 = 196$ transducers operating at 40 kHz with an effective ultrasonic focusing region which is about 90 degrees wide and up to 50cm far. The Ultrahaptics device comes with an API (written in C++ and C#) which has (x, y, z) position coordinates as one of its main inputs needed to generate an amplitude modulated focal point, thereby making the haptification of HGs straight forward as long as they are *i)* detected by the LEAP Motion, and *ii)* within the focusing region of the Ultrahaptics device. Together, they define the IVIS *effective interaction region*.

Processing and Connectivity

The LEAP Motion, the TOUCH[®] Development Kit, the fan, the speakers, and the monitor are all centrally controlled by a Next Unit of Computing (NUC) - a small-form-factor personal computer designed by Intel. This has 3 USB ports, 1 HDMI graphics output, a 3.5mm audio jack, 3Mb of RAM, and an Intel® Core™ i3-7100U Processor, amongst many other features which are not important for our demo. Other processing and connectivity options with similar or better specifications would work equally well. We decided to use a NUC due to its small and portable form factor and affordability.

Physical Prototype Design

Demo Requirements

When designing the prototype physical arrangement of the demo (shown in Figure 3), several aspects were considered including the position of the IVIS relative to the steering wheel, the arrangement of the LEAP Motion relative to the Ultrahaptics device, and the position of the Ultrahaptics device such that the haptified HG effective interaction region is user friendly, intuitive,

and within the peripheral vision region of the driver. Notice that the LEAP Motion and the Ultrahaptics device are co-planar on a kind-of protruding tray. We envisage that this tray could be retractable as to save space or adjustable for different driver seat positions. Moreover, the interaction region should not be near other physical obstacles (e.g., the gear stick) and should not be too high to confuse other drivers with hand waving gestures. Finally, the demo should be transportable (e.g., as oversized cabin baggage).

For demonstration purposes, an 11-inch monitor is included, along with two small fans and two small speakers. In our demo, the user can control the music volume and the fan speed using HGs with their right hand (since the demo is for left-hand drive). Once the 3D design was finalized leaving room for all wiring and fittings at the back of the panel, a custom plastic mold was cast and spray coated in matt black, and all the components assembled and fitted. The Ultrahaptics device was also covered by a woven hydrophobic material which causes negligible signal attenuation yet ensures that the device remains uncontaminated.

Weight, Dimensions and Power Requirements

The whole enclosure weighs 9.5kg. The Ultrahaptics device is just 500g and the LEAP Motion 45g. The prototype is meant to visually approximate a typical car dashboard. Therefore, the demo dimensions occupy a region of $1 \times 0.5 \times 0.5$ meters and sits nicely on a flat table top. A user can interact with the demo while standing directly in front of it. Several other spectators can be standing in the vicinity of the user without affecting the demo operation. The demo requires one 240V power socket for each of its 5 components: NUC, fans, speakers, monitor, Ultrahaptics device. The LEAP Motion is USB powered.

User Interface

The physical prototype setup was designed with the final user software interface in mind. The main goal of the developed IVIS demo was to make the user's interaction as simple and as efficient as possible with a very short and gentle learning curve. Therefore, HGs, and IVIS functions were chosen as to minimize the interaction region and simplify the task at hand.

Dashboard Control

For simplicity, only two dashboard control options were considered for this demo, however others can easily be added retrospectively. The two presented here are volume control, and fan speed control. Each control option is independent, has a different but similar visual representation on the monitor, and can have different HG interaction control methods. In our demo, the user interface menu structure is a simple two level binary tree shown in Figure 5. In more advanced implementations, further breadth and depth options can easily be conceived however one must also consider the trade-off between these [2]. In addition, HG shortcuts could be pre-programmed or be left as user re-programmable. Such options are beyond the scope of the current work.

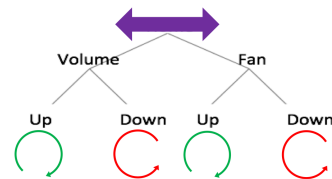


Figure 5: Tree structure of the IVIS menu prototype and HG control schematic color coded to match that of Figure 6.

Hand Gestures

HGs are used to navigate the menu. As previously

mentioned, we wanted to use the LEAP Motion in the most easy and effective way. We therefore designed HGs which utilize the position, orientation, and velocity of the center of the palm. Further, due to the similarity of the two main branches of the menu tree (Up/Down) we concluded that only two gestures were required and their mirror opposites described below.

1. Fan/Volume Up/Down Control: A clockwise or anti-clockwise circular motion of the palm will increase or decrease the control levels respectively in a linear way. The circular motion is detected by directly from the LEAP Motion API which tracks the motion of the center of the palm. The hand speed and circle size are not considered and neither is the distance of the hand from the LEAP Motion. Smaller circles at approximately 20cm above the LEAP and of 10cm diameter tend to work best.
2. Switching modes: With the palm tilted perpendicular to the LEAP, a swiping motion left-to-right or right-to-left will switch between the two main menu functions. Again, the movement of the center of the palm is recorder over a short period and compared to a straight line which is parallel to the LEAP. Since this is a binary event (switch or don't switch) the string length used is restricted to half a second.

The two HGs described above are function associated HGs, that is they are functional gestures that use the natural actions of the hand to associate and provide a cognitive control link between the computer interface and the human user for the function being performed [12]. In addition to these two HGs, we can also recognize a pre-emptive HG when the user's hand is moving towards the interaction region and the

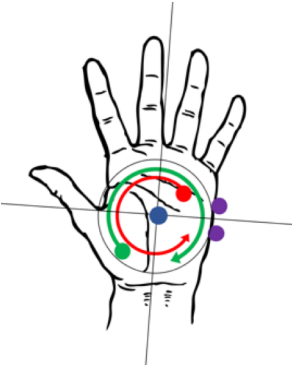


Figure 6: Graphical representation of the hand and the different haptic feedback effects. The blue focal point is at the center of the palm and corresponds to the pre-emptive HG. The green and red focal points move in a circle and correspond to the Up/Down functional HG. Finally, the two purple focal points are located at the middle of the knife-edge of the hand and correspond to the Switching functional HG.

detection of the hand is used to pre-empt driver's intent to operate a particular function.

In more advanced scenarios other hand and finger gestures can be programmed to control the IVIS using the Ultrahaptics API. Already, the LEAP Motion API can track individual finger positions, pointing directions, as well as information about the metacarpal bones that connect the fingers to the base of the hand. As machine vision improves, it is possible that advanced, more complex HGs will be used for IVISs. These however should be intuitive to the user and difficult to confuse with unintentional hand movements.

Ultrasonic Haptic Feedback

Following the classifications of HGs by Pickering et al. [12] we have haptified in our demo two HG application domains. Namely, we have designed one haptic effect for pre-emptive HGs, and two haptic effects for function associated HGs shown in Figure 6.

Pre-emptive HGs: Using the LEAP Motion and Ultrahaptics APIs, we have designed a single Dynamic Focal Point which *clamps* itself onto the driver's palm once the LEAP detects an open palm within the effective interaction region. The driver is therefore haptically informed by the IVIS that her hand has been detected, and is now ready to receive instructions. The breakthrough here is that this process can occur with almost no additional visual demand to the driver. Importantly, the medium by which this haptic feedback is delivered is orthogonal to any other ongoing or interfering audible or visual process.

Function associated HGs:

1. Up/Down control: We have chosen to match the circular HG with a rotating focal point that traces out a small circle that is parallel to the LEAP, the

circle being of radius 2.5cm and centered at the user's palm and moving in the same direction as the HG, i.e., clockwise/anti-clockwise for Up/Down functions. Therefore, the user receives haptic feedback on her hand which is directly correlated to the HG being performed. In the demo, the volume of a music track and the fan speed are also actuated therefore providing the user with additional functional feedback.

2. Switch: we have chosen to create two focal points at the lower edge of the user's hand during the swiping HG. The location of the two focal points is obtained via the LEAP API which calculates the palm center, orientation, and size. This information can be used to locate the knife-edge of the hand. The two focal points are only generated a short bursts during the swiping motion of the hand.

In more advanced scenarios other hand and finger gestures can be haptified to control the IVIS. These may include more focal points, moving or being static, pulsing periodically on and off or at random. As with the HGs, haptic feedback should be meaningful to the user and difficult to confuse. It is therefore the system designer's responsibility to make the technology more seamless, reliable, and natural.

Conclusion

We have presented the world's first prototype demo for a haptified hand gesture (HG) controlled in-vehicle infotainment system (IVIS) which uses focused ultrasound to deliver haptic feedback directly onto the user's hand. The demo has for now limited functions (Volume and Fan speed) however these can easily be expanded. In this extended abstract, we have discussed the main benefits and considerations which

we have attended to whilst designing the physical and software interface of the IVIS as well as the haptified HG interaction techniques. Despite the results demonstrated herein, much research is still needed to understand how to best design haptified HG controlled interfaces for IVISs in order to tackle the many challenges faced by the automotive UI community both from a safety and user experience perspective.

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